

CSCI212 Lab — Quiz #2 (solutions)

1. In Java, Write a static function titled *computeAverageRandomNumber()* under the following conditions:

- Takes two integer parameters *n* and *m*.
- Initializes an **integer array** of size *n*, denoted *randNumArr*.
- Populates *randNumArr* with *n* randomly generated integers between the range **0** and *m* **inclusive**.
- Compute and return the average of *randNumArr*

```
1 public static double computeAverageRandomNumber(int n, int m) {
2     int[] randNumArr = new int[n];
3     double avg = 0;
4     Random rand = new Random();
5     for(int i = 0; i < n; i++) {
6         randNumArr[i] = rand.nextInt(m+1);
7         avg += randNumArr[i];
8     }
9     return avg / n;
10 }
```

2. Suppose you have a 2-D array (a matrix) of integers with dimensions **3x3** (rows x cols), denoted as:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

In linear algebra, the **trace** of a matrix (e.g. *A*) denoted *tr(A)*, is the sum over the diagonal entries of the matrix. i.e. $tr(A) = a_{11} + a_{22} + a_{33}$. In Java, write a static function titled *trace()* that returns the trace of an **nxn** integer array denoted *A*.

IMPORTANT: you are computing the *trace* of an arbitrarily sized **nxn** square array (not fixed to dimensions **3x3**)!

```
1 public static int trace(int[][] A) {
2     int sum = 0;
3     for(int i = 0; i < A.length; i++) sum += A[i][i];
4     return sum;
5 }
```

3. In Java, write a static function titled *totalCharacterCount()* under the following conditions:

- Asks the user to input **10 words** via **System.out.println()**.
- Reads input via **Scanner** object.
- Stores user input into an appropriate array.
- Computes the total number of characters in each word from the array, storing the total count into a variable denoted *characterCount*.
- Returns *characterCount*.

```
1 Scanner scanner = new Scanner(System.in);
2   System.out.print("Enter 10 words: ");
3   String[] words = new String[10];
4   int characterCount = 0;
5   for(int i = 0; i < 10; i++) {
6       words[i] = scanner.next();
7       characterCount += words[i].length();
8   }
9   return characterCount;
10 }
```